

INCIDENT #4471 • STATUS: UNRESOLVED

> searching knowledge base... 0 verified matches on first pass

Institutional knowledge is lost after every incident.

Engineers resolve incidents, document what they learned, and move on. Months later, a different engineer hits the same problem — but can't find the answer that already exists somewhere in the organization, written a different way.

PROBLEM

The same mistake, repeated

After every incident, engineers document root cause, learning, and resolution — usually in a Jira ticket or RCA. The knowledge exists. But when a new incident arrives:

- Past tickets are searched only by title or keyword
- The same failure gets described in inconsistent language
- No system recognizes “this is technically the same problem” across different wording

“The answer already existed. The engineer just couldn't find it.”

Why existing search falls short



Title-only search

Two engineers describe the same failure differently — neither finds the other's ticket.



Inconsistent wording

Root cause, symptoms, and resolution are written in free text with no standard vocabulary.



Isolated knowledge

Learning stays locked in a single ticket, never compounding into organizational memory.

Result: repeat outages caused by problems the organization has already solved.

SOLUTION

An AI system that acts as institutional memory

Automatically finds technically similar past incidents regardless of wording, surfaces verified root cause and resolution, and recommends validation steps before an engineer acts.

1

Learn from History

Continuously ingests resolved incidents, RCAs, and MOPs into searchable institutional memory.



2

Fingerprint Incident

Builds a technical fingerprint — error signature, command signature, node/version, MOP step.



3

Recommend & Hand Off

Surfaces the closest verified matches and hands the decision to the engineer.

The AI never executes changes on its own — it surfaces verified history, the engineer decides.

The matching engine

Three complementary search methods run in parallel, then results are combined and reranked so only the most relevant matches surface.



What the engineer sees

New incident: "BGP session flapping on edge router after firmware push"

INC-203

94% match

Root cause: MTU mismatch introduced by firmware 14.2 default reset on edge routers. Fix: Pin MTU value in post-upgrade config template.

INC-119

71% match

Root cause: BGP timer defaults reverted after vendor patch. Fix: Reapply custom keepalive/hold timers via automation.

AI RECOMMENDATION

Matches INC-203 — firmware reset MTU to vendor default.

Apply the INC-203 MTU override before re-enabling the session.

Engineer reviews and confirms before any change is applied.

The continuous learning loop

Finds similar past incidents, surfaces verified learnings, and helps engineers avoid repeating the same mistakes.

STEP 1

Historical Data

Jira, MOPs, RCAs, records, logs.

STEP 2

Knowledge Base

Structured data + vector embeddings.

STEP 3

New Incident

Error, symptoms, commands, environment.

STEP 4

Fingerprint

Error/command signature, node, MOP step.

STEP 5

Hybrid RAG Search

Semantic + keyword + metadata match.

STEP 6

Ranked Matches

Similar incidents ranked by confidence.

STEP 7

AI Recommendation

Root cause, learning, resolution, why.

STEP 8

Engineer Validates

Reviews, confirms, then acts.

STEP 9

New Learning

Verified fix written back for reuse.

Continuous learning loop — the system keeps improving with every new incident.

A new standard for incident learning

Old Way	This Solution
Search by ticket title	Search by technical fingerprint
Misses reworded incidents	Finds matches regardless of wording
Manual digging through old tickets	Automatic surfacing of relevant history
Knowledge locked in one ticket	Knowledge compounds over time
AI could hallucinate a fix	AI only cites verified past evidence

KEY DESIGN PRINCIPLE

The AI never invents historical facts — it only surfaces and summarizes what was already verified and documented. A human always validates before any production action.